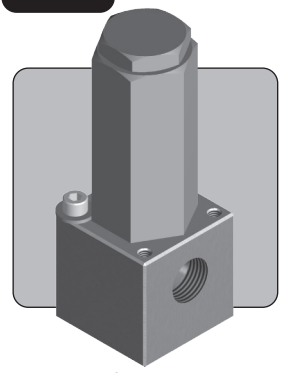
RL50

RELIEF VALVE

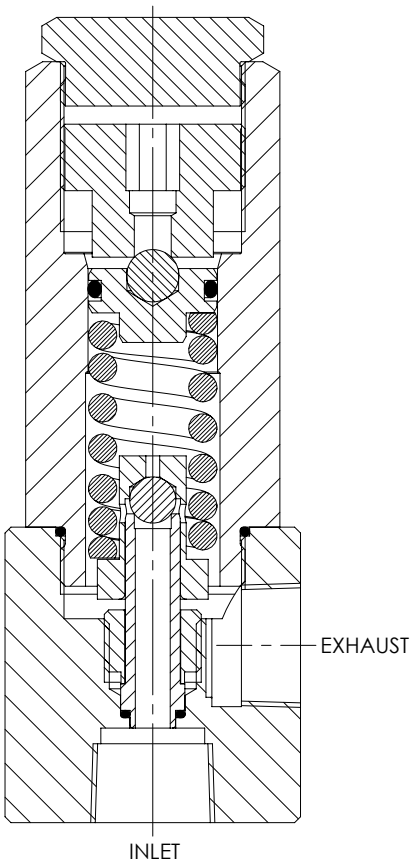
1/2″ UP TO 1,034 BAR 15,000 PSI



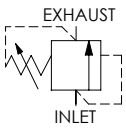
The RL50 is a 1/2″ ported relief valve. Designed for emergency relief applications, it may be used to protect systems up to 1,034 bar (15,000 psi).

The valve is a proportional metering style relief valve, and is approved as a safety accessory to category IV of the Pressure Equipment Directive, enabling its use in the most demanding applications.

- Approved to category IV of the Pressure Equipment Directive (2014/68/EU)
- Available preset via internal adjustment or with an external handwheel adjustment option
- Wire locking and tagging available on request
- Minimal leakage at 90% of set pressure
- Reseat within 20% of set pressure
- Repeatabily within ±5% of set pressure
- Subsea variants available, contact us for advice
- Suitable for use with mineral oils and water glycols, with options for water and sea water duty
- Suitable for many other media, contact us for advice
- Various spring ranges available



Specifications

BASIC MODEL NUMBER	RL50
SYMBOL	
MAX WORKING PRESSURE	1,034 bar (15,000 psi)
MAX OUTLET PORT PRESSURE	69 bar (1,000 psi) Note: Any outlet port pressure is directly additive to the set pressure
ORIFICE SIZE	Ø6.0mm (Ø0.236″) - 1.6K & 5K Ø4.1mm (Ø0.161″) - 15K
FLUID	Liquids only See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	1/2″
WEIGHT	1.75 kg (3.9 lb)

Specifications may change without notice

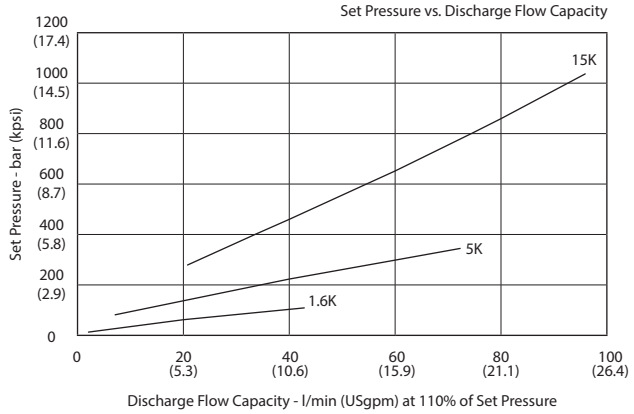
Materials

Externally Exposed Parts: 316 stainless steel on all pressure retaining components, and 300 series stainless steels on non-critical parts.

Internally Wetted Parts: 316, 302 and 431 stainless steel. The seat and ball are 440C stainless steel and silicon nitride respectively. The water duty version contains a 17-4 PH stainless steel seat. The sea water duty valve uses only 316 stainless steel, inconel and silicon nitride in wetted areas. Note that use with water above 100°C is not recommended.

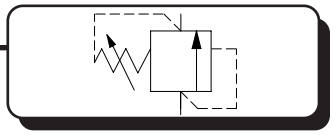
⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

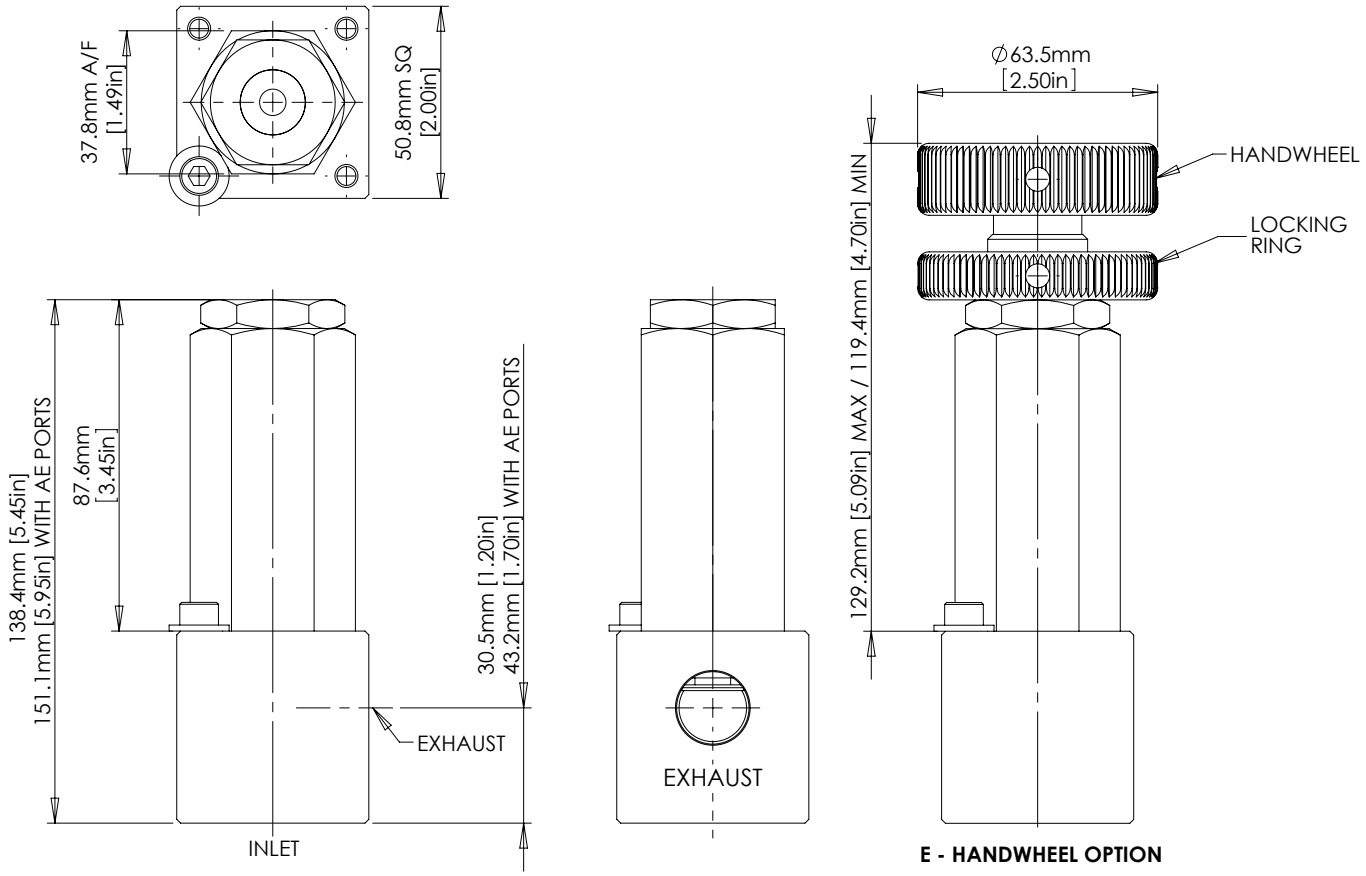
Typical Performance



Typical performance based on water
Contact us with system details for an application specific calculation

Installation Diagram





E - HANDWHEEL OPTION

Product Selector

RL50 - 56AE - 50P - E - WD - NBR - 1.6K

VALVE TYPE	INLET PORT OPTIONS	EXHAUST PORT OPTIONS	VALVE OPTIONS	WORKING FLUID OPTIONS	SEAL OPTIONS	SET PRESSURE RANGE
		Leave blank for P and N inlet options	Leave blank if none required	Leave blank if none required	Leave blank for Viton® seals -10˚C to +120˚C	
RL50	**P** 1/2" BSPP female inlet and exhaust	**50P** 1/2" BSPP female exhaust	**E** Handwheel (external) adjustment	**WD** Water duty	**NBR** Nitrile seals -30˚C to +105˚C	**1.6K** 13.8 - 110 bar (200 - 1,600 psi)
	N 1/2" NPT female inlet and exhaust	**50N** 1/2" NPT female exhaust	⚠ Exposure to high and low fluid temperatures will influence handle temperature. Operator safety must be considered	**SWD** Sea water duty	**KLZ** Kalrez® seals -25˚C to +200˚C	**5K** 69 - 345 bar (1,000 - 5,000 psi)
	56AE 3/8" OD medium pressure tube inlet				**HNBR** Hydrogenated Nitrile seals -20˚C to +150˚C	**15K** 276 - 1,034 bar (4,000 - 15,000 psi)
	81AE 9/16" OD medium pressure tube inlet				Further seal options are available on request	Advise us of set pressure if factory pre-setting is required

Repair and seal kits are available on request

For a seal kit add SK at the end of the model code and for a repair kit add RK